

ELECTIVE COURSE-4.1

URA417: VEHICLE SAFETY ENGINEERING				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: The purpose of learning this course is to broaden the knowledge of crash testing - regulations and legislations, different vehicle body design towards safety, active and passive safety systems, driver and occupant biomechanics and to acquire experience by carrying out virtual collision model simulation.				
Unit 1: Safety Testing and Legislation Need of the safety aspects in vehicle design and development - Necessity behind the safety testing and legislation (FMVSS - etc.) - Types of safety systems - New Car Assessment Program – Procedure - rules and regulations - rating methods - Behavior of specific body structures in crash testing - Photographic analysis of impact tests - Pedestrian Impact - Crash test dummies. Flipped Classroom: Crash test – frontal impact - rear impact - side impact - and Impact with rebound - movable barrier tests; Roll over crash tests Unit 2: Vehicle body design towards safety Design of the body for safety - energy equation - powertrain component's location (engine - motor - battery pack - etc.) - deceleration of vehicle inside passenger compartment - deceleration on impact with stationary and movable obstacle - Exterior safety - interior safety - deformation behavior of vehicle body - Balance of stiffness and toughness characteristics and energy absorption characteristics of vehicle structures. Flipped Classroom: concept of crumple zone - safety sandwich construction - Optimization of vehicle structures for crash worthiness Unit 3: Biomechanics and Occupant safety and Contemporary Issues Biomechanics - Human impact tolerance - Injury tolerance limits - Severity Index - Study of comparative tolerance - Application of Trauma for analysis of crash injuries - External injuries - Internal injuries – Concussion - Spinal and Chest injuries - Occupant protection – Head - Chest Neck				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

- Pelvic - Leg and Knee - Pedestrian protection - Criteria for Injury in safety aspects - Importance of Ergonomics in Automotive safety - Locations of controls – Driver seat design.

Flipped Classroom: Autonomous vehicle and its safety challenges,

Industry Expert Lecture: HCV Crash and Traffic Safety – By Volvo

Unit 4: Passive and Active Safety System

Survival space requirements - restraints systems - Seat belts – retractor - pre - tensioner - load limiter; Head restraints; Air bags – types - working sequence - activation sensors; Impact protection from steering controls - Design of seats for safety - types of seats used in automobiles. Bumpers - Use of energy absorbing systems - crash box - damageability criteria in bumper designs - Hinges - latches - central locking - etc.

Unit 5: Vehicle Collision Models and Computer support for development of Safety systems

Impulsive models - Perfect plastic impact - Perfect elastic impact - Co - efficient of restitution - Central Collision - central head on collision - oblique collision - collision against fixed obstacle - Non central Collision - non - central head on offset collision - Kelvin's theorem - Application of relative motion - Change in vehicle speeds - Total crush energy - Vehicle individual crush energies - Crash severity assessment. Problems involving vehicle collisions

Flipped Classroom: Numerical tools – Basics - Component level testing and analysis; Total vehicle crash computation – Dynamic vehicle simulation tests – Frontal collision - Lateral collision - Rear end collision and Roll over - Occupant and restraint system simulation – Seats - seat belt - air bag - head restraint - ADAS system performance improvement through simulation.

Course Learning Objectives (CLO)

The students will be able to:

1. Demonstrate the knowledge about safety and vehicle structural crashworthiness
2. Analyze the importance of human bio - mechanics in vehicle safety design
3. Analyze the performance of active and passive safety systems.
4. Analyze and suggest suitable collision models for vehicle safety

Text Books

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.